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Training methods to improve vertical jump performance

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Aim. **This study aims to review the main methods used to improve vertical jump performance (VJP).**
Methods. **Although many training routines have been proposed, these can be grouped into four main categories: plyometric training (PT), weight training (WT), whole body vibration training (VT) and electromyostimulation training (ET). PT enhances muscular force, the rate of force development (RFD), muscular power, muscle contraction velocity, cross-sectional area (CSA), muscle stiffness allowing greater storage and release of elastic energy. WT improve muscular force, velocity, power output, and RFD during jumping on a force plate, muscle hypertrophy and neural adaptations. One of the most effective methods to improve VJP is the combination of PT with WT, which takes advantage of the enhancement of maximal dynamic force through WT and the positive effects of PT on speed and force of muscle contraction through its specific effect on type II fibers.**
Results. **Some authors have found an increase in VJP with the use of VT while other did not see such an effect. However, it remains unknown by which mechanisms VT could enhance VJP. ET has been shown to elicit muscle hypertrophy. The VJP may be improved when ET is applied concomitantly with PT or practice of sports.**
Conclusion. **In summary, scientific evidence suggests that the best way to improve VJP is through the combination of PT with WT. Further research is needed to establish if better results are possible by more complex strategies.**

KEY WORDS: Plyometric exercise - Weight lifting - Exercise.

In many sports is important to improve vertical jump performance (VJP). Vertical jump capacity (VJC) is associated with success in many sports.[1, 2] The VJC depends in part on the lower extremities muscle power and has been used as a standard tests of power performance,[3] and to estimate the composition of the muscular fibers.[4]

To improve VJP a greater vertical velocity at take-off is required,[5] which may be achieved by a higher contraction velocity and/or muscle force of the extensor muscles of the trunk, hip and lower extremities. Several studies have examined the effects of a wide variety of training methods on VJP, which include: plyometric training (PT),[6-47] weight training (WT),[6-10, 12, 15, 20, 44-69] whole body vibration training (VT) [62, 66, 70-86] and eletromyostimulation training (ET).[18, 87-95] All these methods have been studied alone or in combination with other methods.

Therefore, the aim of this study is describe the main training methods applied to improve VJP.

Main training methods to improve vertical jump performance

VJP can be improved by PT, WT, VT or ET.[17, 65, 83, 95] The rationale for the use of these methods to improve VJP is determined by the close relationship observed between the maximum dynamic force of the lower extremities and the maximum height achieved during vertical jumping.[96] Subjects with higher isometric force and/or enhanced rate of force development

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(RFD) during knee extension have also better VJP.[97] A positive correlation (r=0.81) between maximal leg extension isometric force and vertical jump height have been found.[98] Similarly, the activation of the femoral quadriceps contributes 50% of the work applied in the vertical jump.[99] It has been shown that better jumpers achieved greater joint moments, power and work done at the ankle, knee and hip, and as a result they reached higher height in the countermovement jump (CMJ) with and without arm swing. The authors concluded that the superior performance of the better jumpers was due to greater muscle capability in terms of strength and RFD in all lower limb joints rather than to technique, which differed less noticeably between the groups, and had less of an effect.[100]

Plyometric training

PT is the most commonly used method to improve VJP and leg muscle power between coaches and researchers.[101] The squat jump (SJ), CMJ and drop jump (DJ) are the most known plyometric exercises. The last two are characterized by a stretch shortening cycle (SSC) movement. During the SSC two difference phases can be observed: a first eccentric contraction followed immediately by a fast concentric contraction (shortening). In the eccentric contraction the muscles contract to perform a rapid deceleration to brake rapidly the downwards movement of body mass centre. This involves the lengthening of the contracting agonist musculature (stretching). During the fast concentric contraction the centre of body mass is accelerated in the upward direction. Three main mechanisms have been proposed to explain how a SSC allows for a higher production of force at a faster velocity: mechanical potentiation, higher force application time and post-activation potentiation (PAP). In the mechanical potentiation is believed that the SSC evokes the elastic properties of the muscle fibers after a quick stretching of the tendon-muscle structure, which allows muscles to store energy in the series elastic elements during the eccentric phase. This energy is released quickly and if the stretching is immediately followed by a concentric muscular action, part of this energy contributes to enhance the power generated during the concentric phase.[102, 103] The mechanisms of higher force application time is used because during a vertical jump the time available to generate tension is rather short (in general less than 0.4 s),[104] substantially less than required to attain the maximal isometric force. The peak force achieved during the SSC may be enhanced by increasing the time available for muscle contraction. Svantesson *et al.* showed that the level of force developed during a concentric muscle contraction is enhanced by a preceding eccentric contraction, which allows for storage of elastic energy but also by a preceding isometric contraction which does not allow for a significant accumulation of elastic energy. However, the degree of potentiation was larger with eccentric than with isometric preceding action, regardless of movement velocity. The study showed that the main reason for larger concentric torque values after a preceding muscle action is that time is sufficient for maximal muscle tension development.[105] Another mechanism is the PAP, it is know that peak force, the RFD and the time needed for force to attain maximal values are enhanced following a sustained maximal voluntary contraction (MVC),[106] muscle performance can be improved when preceded by contractile activity.[107] This phenomenon was noted first by Ranke,[108] who described that with stimuli uniform in strength the later twitch contractions were stronger than the first.

The PT may stimulate muscle hypertrophy, particularly of type II fibers as observed in volleyball players.[109] This could be related to the enhanced recruitment of FT motor units during eccentric muscles contractions.[110] Increased muscle stiffness has the advantage of allowing greater storage and release of elastic energy.[111] In addition, a positive correlation has been described between RFD and connective tissue stiffness measured in vivo, as well as between maximal jumping height and the stiffness of the force-transmitting tissues.[111] Potteiger *et al.* observed a significant increase in fiber area for type I and II after 8-week of PT. They suggested that the increased in VJP and power output could be related to the increased in muscle fiber size.[35] It has been suggested that part of the improvement vertical jump following PT depends also on neuromuscular adaptations such as the increased inhibition of antagonist muscles after training, better co-contraction or increased activation of synergistic muscles, reduction of neural inhibitory mechanisms, such as a Golgi tendon receptors discharge, and increased agonist motoneuron excitability and synchronization.[41, 112-118]

Effects of plyometric training on vertical jump performance

Although, some studies were contradictory,[42, 46, 61, 119] the majority of investigations have shown that PT improved VJP (Table I) in untrained subjects 5% to 35% [6-9, 11, 14, 15, 17, 20, 21, 24, 25, 27-29, 31, 32, 35, 36, 41, 44] or trained subjects 6 to 13%.[10, 22, 30, 37, 38, 45]

The highest improvement in VJP in untrained people has been observed by Kotzamanidis [24] with a 35% of increased in SJ height after ten weeks of PT, two days per week, one reason that can explain the high increment could be a low status training of the prepubescent with 11 years old, however it should not be the only one because Diallo *et al.*[14] observed a lower increment (7%) in SJ height after also ten weeks of PT in prepubescent boys aged 12 years training with 3 days of training per week. The lowers improvement in VJP in untrained people was observed by Potteiger *et al.*[35] with and increased of 5% in VJP.

Higher improvements in vertical jumps are often seen with higher intensity of training, Gehri *et al.*[17] observed that PT using DJ reached higher performance than using CMJ in VJP, the improvement in SJ, CMJ and DJ was 14%, 8% and 11% respectively in the DJ group, while the respective improvements in the group trained with CMJ exercises were 7%, 5% and 9% respectively. Matavulj *et al.*[30] reported that after 6-week of PT the group who trained DJ (100 cm) improved VJP a 5.6%, more than the group who trained DJ (50 cm) that improved VJP a 4.8%.

In order to reduce the stress due to the impact associated with plyometric exercises, some studies observed that the effect of PT in an aquatic environment reduces the risk of injuries due to the density and viscosity of the water.[29, 39] Stemm and Jacobson [39] observed that 6 weeks of plyometric exercises, twice a week, doing 3 sets of 15 squat jumps, side hops, and knee-tuck jumps, by one group in land and the another group in aquatic conditions, both groups significantly outperformed the control group in the vertical jump, without significant difference in VJP between aquatic and land plyometric groups. The control group had no training. Martel *et al.*[29] observed that the combination of aquatic plyometric group and volleyball training improved significantly VJP by 8% from week 4 to week 6 compared to the control group that performed flexibility exercises combined with volleyball training.

Weight training

WT is a method based on lifting loads, when WT uses loads approximately from 80 to 100% of maximal dynamic strength (1RM), which are lifted approximately 1-8 times, it is called heavy load WT or traditional WT and has been proved to elicits remarkable increases of 1RM through muscle hypertrophy and neural mechanisms.[120] When the intensity of the loads lifted are around 30-50% of 1RM the movement is more explosive and have been found to be very effective to improve mechanical power in movements that require explosiveness.[45, 63] The light load WT has been adapted to incorporate more dynamic and explosive movements to promote power development, with light loads large accelerations can be achieved at the beginning of the concentric phase of the movement, however, the load must be stopped at the end of the range of motion. The deceleration phase accounted for 24% of the concentric phase of the bench press when executed with a maximal load and increased to 52% when the bench press was performed at 81% of 1RM. [121] To avoid this problem the load should be propelled *ballistic training* [122] or in case of squat exercises a weighted jump squat technique should be used, to accelerate all the way through the movement, the Plyometric Power System let do it and avoid injury using an eccentric brake system that remove 75% of the weight of the bar on the eccentric phase. [123]

WT elicits morphological, neural and hormonal adaptations that enhance 1RM and RDF.[124, 125] Morphological adaptations are changes in the whole muscle size, fibre type and myosin heavy chain (MHC), muscle fibre hypertrophy, hyperplasia and myofibrillar proliferation.[125] Neural adaptations include increases in motor unit activation, firing frequency, synchronization of motor units, coordination, agonist activation and decreases in antagonist co-activation.[120, 125] WT also increases the concentration of some anabolic hormones, such as testosterone, growth hormone, insulin and insulin-like growth factor-1. [124]

Effects of weight training on vertical jump performance

Most of studies have found that WT improved VJP (Table II) in untrained subjects from 2 to 25%,[6-9, 15,

TABLE I.—*Improvement in vertical jump performance with plyometric training (PT).*

References	Training					
	Type		We	Se	Sets	Rep
Adams *et al.*[6]	PT	DJ(51-114)+jumps	6	2		
Anderst *et al.*[7]	PT	Jumps	12	3		
Arabatzi *et al.*[8]	PT	Jumps	8	3	4-6	6
Bauer *et al.*[9]	HG-PT		10	3		
Brown *et al.*[11]	PT	DJ(45)	12	3	3	10
Diallo *et al.*[14]	PT	DJ(30-40)+jumps	10	3		
Fatouros *et al.*[15]	PT	DJ(30-80)+jumps	12	3		
Gehri *et al.*[17]	PT	CMJ	12	2	2-4	8
		DJ(40)	12	2	2-4	8
Holcomb *et al.*[20]	PT	DJ(40-60)	8	3	9	8
Impellizzeri *et al.*[21]	PT+SoT	On sand	4	3	3-25	5-15
		On grass	4	3	3-25	5-15
Kotzamanidis [24]	PT	DJ(10-30)+jumps	10	2		
Kyrolainen *et al.*[25]	PT	DJ(20-70)+jumps	15	2		
Malisoux *et al.*[27]	PT	DJ(40)+jumps	8	3		
Markovic *et al.*[28]	PT	DJ(40)+hurdles	10	3	4-10	10
Martel *et al.*[29]	PT-A+VoT	DJ+jumps	6	2		
McClenton *et al.*[31]	PT	DJ(50-100)	6	2	2-4	4-10
Meylan *et al.*[32]	PT+SoT	Jumps	8	2	2-4	6-12
Potteiger *et al.*[35]	PT	DJ(40)+jumps	8	3	1-8	4-10
	PT+AE	DJ(40)+jumps	8	3	1-8	4-10
Rubley *et al.*[36]	PT+SoT	DJ(25)+jumps	12	1	1-4	5-20
Toumi *et al.*[41]	PT-ISO	0.4 m/s	8	4	6	10
		0.2 m/s				
Wilson *et al.*[44]	PT-UL	DJ(20-70)+Mbt	8	3	1-4	8
Berryman *et al.*[10]	PT+EnT	DJ(20-60)	8	1		
Khlifa *et al.*[22]	PT+BaT		10	2-3	3-25	5-15
	PT-L+BaT		10	2-3	3-25	5-15
Matavulj *et al.*[30]	PT	DJ(50)	6	3	3	10
		DJ(100)	6	3	3	10
Sheppard *et al.*[37]	PT-AJ		5	3	5-7	5
Spurrs *et al.*[38]	PT+EnT	DJ+jumps	6	2-3	2-3	6-12
Wilson *et al.*[45]	PT	DJ(20-80)	10	2	3-6	

HG-PT: hydra gym with plyometrics; SoT: soccer training; PT-A: aquatic plyometric training; VoT: volleyball training; AE: aerobic exercise; PT-ISO: plyometric training isokinetic ergometer; UL: upper and lower body; EnT: Endurance training; BaT: basketball training; PT-L: loaded plyometric training; AJ: assisted jumping; DJ(cm): drop jump height in centimetres; jumps: various types of jumps; Mbt: medicine ball trows; we: weeks; Se: sessions per week; Rep: repetitions; N.: number of subjects; Ge: gender; F: female; M: male; yr: years: SJ: squat jump; CMJ: countermovement jump; DJ: drop jump; V: other kind of jump.

[44, 48, 50-52, 56, 57, 60, 61, 63, 64, 67, 68, 81] and in trained subjects from 3 to 18%.[10, 45, 49, 58, 62, 65, 69]

The best results in enhancing dynamic athletic performance on VJ has been observed using light loads, at high speed. It showed the biggest improvement in trained people in SJ (15%) and CMJ (18%) compared with the improvement in SJ (7%) and CMJ (5%) achieved with heavy loads. [45] However, Lyttle *et al.* [63] observed in untrained subjects that light loads and combined WT and PT have similar effects on jumping performance. But, the combination of WT with PT yielded better results when subjects had to perform SSC activities. This may be caused by the fact that PT is more dynamic than WT and hence may better facilitate the neural and mechanical mechanisms that enhance performance in SSC activities.

Participants				Gain							
				Cm				%			
N	Ge	Type	Yr	SJ	CMJ	DJ	V	SJ	CMJ	DJ	V
		Untrained, students, recreational									
12	M	Intermediate lifters					3.8				
5	M	Recreational athletes					8.4				
9	M	Students	20	4.2	4.6			14.1	14.6		
6	M-F	Students	21				5				10
13	M	Students basketball players	15				7.3				12.5
10	M	Soccer players	12	2	3.4			7.3	11.6		
11	M	Untrained	21				6				11.3
7	M-F	Students	19	1.9	1.7	2.4		6.8	5.4	8.7	
11	M-F		20	3.3	2.1	2.8		13.6	8	10.9	
10	M	Students			6.1				12.4		
19		Amateur soccer players	25	3.5	2.4			10.2	6.5		
18			25		5.5				14.6		
15	M	Prepubescent	11					34.7			
13	M	Recreational active	24			7				23.3	
8	M	Recreationally active	23	3	6			9	13		
30	M	Students physical education	20					6.5	6.3	14.2	
10	F	Students volleyball players	15		3.4				11.1		
10	M-F	Recreationally trained students	21				2.2				10.5
14		Soccer players	13		2.6				7.9		
8	M	Physically active	21				2.7				4.6
11	M		21				3.1				5
10	F	Students soccer players	13				7.4				18.6
12	M	Sedentary	19-23	2.9	5.2			8.5	13.9		
			19-22	3.4	3.5			9.4	8.9		
14	M	Students	20.5				10.4				18
		Trained, Elite, Athletes									
11	M	Moderately-welltrained runners	29		2				6		
9	M	Elite basketball players	24	2.2	3.1			5.8	7		
9	M		23	3.7	5.3			9.9	12.2		
11	M	Elite basketball players	15-16		4.8						
11	M		15-16		5.6						
7	M	Elite volleyball players	18		2.7						
8	M	Distance runners trained	25		5				13.2		
13	M	Trained	22		3.7				10.3		

Similarly, Newton *et al.*[65] have shown that 8-week of *ballistic training* improved the VJP in elite volleyball players by enhancing force, velocity, power, and RFD during jumping on a force plate. In addition, training with heavy loads may also enhance the speed of movements [126] probably because it improves 1RM. However, WT with short recovery periods between series appears less efficient to enhance VJP.[127]

Combined plyometric training and weight training

Jumping height performance depends on take off velocity which is determined by the ability of muscles to both achieve a high level of force in a rather short time. The combination of WT with PT aims at taken advantage of the enhancement 1RM through

TABLE II.—*Improvement in vertical jump performance with weight training (WT).*

Referentes	Training					Participants			
	Type	We	Se	Sets	Rep	N.	Ge	Type	Yr
								Untrained, Students, Recreational	
Adams *et al.*[6]	WT	6	2	1-4	2-8	12	M	Intermediate lifters	
Anderst *et al.*[7]	WT	12	3			7	M	Recreational athletes	
Arabatzi *et al.*[8]	OL	8	3	4-6	4-6	9	M	Students	20
Augustsson *et al.*[48]	WT-CKC	6	2			11	M-F	Students	25
Bauer *et al.*[9]	WT	10	3	4-7		8	M-F	Students	22
Channell *et al.*[50]	WT-OL	8	3	3-5	3-20	11	M	Students	16
	WT-PL	8	3	3-5	3-20	10	M		16
Chelly *et al.*[51]	WT+SoT	8	2	3	4-2	11	M	Soccer players	17
Christou *et al.*[52]	WT+SoT	16	2	2-3	8-15	9	M	Soccer players	14
Faigenbaum *et al.*[57]	WT	9	2	3	1-4	22	M	Students	14
Fatouros *et al.*[15]	WT	12	3	7		10	M	Untrained	21
Granacher *et al.*[57]	WT-Ba	8	2	4	10	14	M-F	Students	17
Kalapotharakos *et al.*[60]	WT	12	3			9	F	Inactive, sedentary	53-69
Kramer *et al.*[61]	WT+ReT	9	3	4-10	5-20	11	F	Novice and experience	21-22
Lyttle *et al.*[63]	WT-MP	8	2	2-6	8	11	M	Untrained	24
Moore *et al.*[64]	OL+WT	12	3	3	6	8	M-F	Collegiate soccer players-Untrained	20
Roelants *et al.*[81]	WT	24	3			20	F	Untrained	64
Tricoli *et al.*[67]	WT	8	3	3-6	4-6	12	M	Students	22
Weltman *et al.*[68]	WT-H	14	3			16	M	Boys	8.2
Wilson *et al.*[44]	WT-UL	8	3	3-6	6-10	14	M	Students	22.4
								Trained, Elite, Athletes	
Baker *et al.*[49]	WT-NPM	12	3	5-3		9	M	Experienced athletes	19
	WT-LPM	12	3	5-3		8	M		20
	WT-UPM	12	3	5-3		5	M		21
Berryman *et al.*[10]	WT+EnT	8	1	3-6	8	12	M	Moderately-welltrained runners	31
Harris *et al.*[58]	HP	9	4	5	5	16	M	Trained	19
	HF + HP	9	4	5	5	13	M		20
Kvorning *et al.* [62]	WT	9	1-3	3-6	8	9	M	Moderately trained	24
Newton *et al.* [65]	WT-Ba	8	2	6	6	8	M	Trained	19
Wilson *et al.* [45]	WT	10	2	3-6	6-10	15	M	Trained	22
	WT-MP	10	2	3-6	6-10	13	M		24
Wong *et al.* [69]	WT+EnT	8	2	4	6	20	M	Professional soccer players	25

OL: olympic lifts; CKC: closed kinetic chain; PL: power lifts; SoT: Soccer training; Ba: ballistic training; ReT: rowing ergometer training; MP: maximal power training; H: hydraulic; UL: upper and lower body; NPM: non periodized methods; LPM: linear periodized method; UPM: undulating periodized method; EnT: endurance training; HP: high power; HF: high force; We: Weeks; Se: sessions per week; Rep: repetitions; N.: number of subjects: Ge: gender: F: female; M: male; Yr: years; %RM: percent of maximum load; MIF: maximum isometric force; RM: repetition maximum load; SJ: Squat jump; CMJ: countermovement jump; DJ: drop jump; V: other kind of jump.

WT [128] and the positive effects of PT on speed and force of muscle contraction through its specific effect on type II fibers.[111] Wilson *et al.*[44] postulated that WT facilitates principally the concentric performance, whereas PT emphasizes the eccentric component and the RFD. Recently, Kubo *et al.*[129] showed that WT improved only concentric jump performance mainly by neural adaptations and muscle hypertrophy, while PT improved concentric and SSC performance attributed to changes in mechanical properties of muscle-tendon complex, rather than neural activations. Although it is rather difficult to change muscle fiber types from slow to fast with strength training,[130] the combination of strength training with ballistic and SSC movements is able to promote an increase in MHC IIa and a decrease in MHC I. [131] In addition, this type of combined training seems to attenuate the reduction in MHC IIx normally seen

Intensity		Gain							
		Cm				i%			
%rm	Rm	Sj	Cmj	Dj	V	Sj	Cmj	DJ	V
50-100					3.3				
					5.5				
75-90	4-6	5.7	5.2			20.3	15		
			4.9				10.2		
60					4.5				8.9
					2.6				4.5
					1.1				2.3
80-90		0.6				2			
55-80		4.5	2.6			16.1	7.9		
	8-15				2.2				5
70-95					5.4				9.3
30-40			2.1				7.7		
80						24.5	21.7		
					2.1				5.6
30		7.1	3.8			18.4	7.5		
					4.2				9
	8-20						12.9		
	4-6	3.7	2.8			9.5	6.6		
					2.2				10.4
					11.2				21.2
	6-8		4.5						
	10-1		2						
	10-3		4.9						
			1.5				4.5		
30% MIF			2.3				3.9		
80-85			1.8				2.9		
	8-10		2.3				7.8		
30-60-80			3.9				5.9		
	6-10	2.4	1.9			6.8	5.1		
30		5	6			15	18		
	6				2.5				3.9

with strength training.[132] Thus, a combination of WT and PT would be a better training strategy to enhance power than either WT or PT alone.[133]

Complex training is a training strategy that combines WT and PT in the same training session.[134] This training method alternates similar WT exercises of high loads, set for set, in the same workout prior to doing specific plyometric exercises to improve dynamic movement. In complex training subjects may perform a set of squats followed by a set of DJs. When two biomechanically similar exercises, one consisting in heavy weight lifting and the other in plyometric exercises (squat jumps, for example), are performed sequentially the training routine is called complex pair.[135] It has been suggested that this training method has an acute ergogenic effect on power and improves the jumping performance [136] because it may elicit PAP.[137] For example, the height reached

TABLE III.—*Improvement in vertical jump performance combining plyometric training (PT) and weight training (WT).*

References	Training					Participants			
	Type	We	Se	Sets	Rep	N	Ge	Type	Yr
								Untrained, Students, Recreational	
Adams *et al.*[6]	PT+WT	6	2			12	M	Intermediate lifters	
Arabatzi *et al.*[8]	WT+PT	8	3			10	M	Students	20
Bauer *et al.*[9]	WT+PT	10	3			7	M-F	Students	25
Blakey *et al.*[124]	PT (DJ 110 cm)+WT	8	3			11	M	Students	18-21
	PT (DJ 40 cm)+WT	8	3			10	M		
	PT (DJ 0 cm)+WT	8	3			10	M		
Fatouros *et al.*[15]	PT+WT	12	3			10	M	Untrained	20
Fowler *et al.*[125]	PT-PeS + WT	3	4			9	M	Students	23
Hakkinen [126]	PT+WT+BaT	22	2-3			10	F	Basketball players	
Holtz *et al.*[127]	PT+WT+BaT	10	3			6		Junior college basketball players	
Hunter *et al.*[128]	PT+WT+FlT	10	2-4			14	M	Untrained	24
	PT+WT		2			11			24
Ingle *et al.*[59]	WT+PT	12	3			26	M	Students	12
Kramer *et al.*[61]	PT+WT+ReT	9	3-4			13	F	Novice and experienced	21-22
Lyttle *et al.*[63]	WT+PT	8	2			11	M	Untrained	24
Mihalik *et al.*[123]	PT+WT	4	2			15	M-F	Volleyball players	20
	CT					16			21
Moore *et al.*[64]	PT+WT	12	3			7	M-F	Collegiate soccer players-Untrained	21
Perez-Gomez *et al.*[113]	PT+WT	6	3			16	M	Students	23
Tricoli *et al.*[67]	PT+WT	8	3			12	M	Students	22
Trzaskoma *et al.*[129]	PT-PeS+WT	3	4			9	M	Students	23
Wong *et al.*[130]	WT+PT	12	2			28	M	Soccer players	14
								Trained, Elite, Athletes	
Maio Alves *et al.*[131]	CCT	6	1			9	M	Elite soccer players	17
			2			8			17
Marques *et al.*[132]	WT+PT+VoT	12	2			10	F	Elite volleyball players	25

PeS: pendulum swing; BaT: basketball training; FlT: flexibility training; ReT: rowing ergometer training; CT: compound training; CCT: complex and contrast training; VoT: volleyball training; We: weeks; Se: sessions per week; Rep: repetitions; N.: number of subjects; Ge: gender; F: female; M: male; Yr: years; SJ: squat jump; CMJ: countermovement jump; DJ: drop jump:; V: other kind of jump.

during loaded CMJ performed after a set of half squat at 5-RM was significantly higher (2.8%) than the same loaded CMJ executed immediately preceding the half-squats.[138] This effect may be observed more easily in subjects with a high level of strength. Chiu *et al.*[139] reported that strength-trained subjects enhanced their power performance for 5 to 18.5 minutes, while recreationally trained individuals showed fatigue in the following 5 minutes after heavy WT.

It has been recently proposed to combine agonist and antagonist muscle exercises into the same power training session to elicit an acute increase in power output in the agonist power exercise.[140] These authors studied a group of 24 college-aged rugby league players who were experienced in combined strength and power training, they were randomly assigned to an experimental or control group. The outcome variable was power output assessed during bench press throws with a 40-kg resistance with the Plyometric Power System training device. After warming up, the control group performed the bench press throws tests 3 minutes apart to determine if any acute augmentation to power output could occur without intervention. The experimental group also performed the bench press throws before and after a set of bench pulls (the antagonistic action to the bench throw). Although the power output for the control group remained unaltered between the two tests, it was increased by 4.7% in the experimental group.[140] Thus, combination of agonist and antago-

Gain							
Cm				%			
Sj	Cmj	Dj	V	Sj	Cmj	DJ	V
			10.7				
4.3	5.2			14.6	15.1		
			4.1				8
			11.1				
			11.4				
			12.2				
			8.6				14.6
	3				6.1		
2.5	1.4			11.5	5.6		
			5.1				7.3
	4.9	3.5			12.7	9	
	2.9	3.7			7.9	10.7	
	1.3				4		
			2.2				5.6
6.7	5.6			16.6	10.6		
	2.7				5		
	4.8				9		
			2.8				7
	2.8				8.3		
	2.3				5.7		
	3				6.2		
	3.3				6		
5.2				12.6			
3.8				9.6			
	1.3				3.8		

nist exercises may acutely increase power output. However, it remains to be determined if such a strategy may lead to a greater adaptation when used systematically during the training sessions.

Compound training is a training strategy that performed WT and PT in different training sessions. Although recommended by many in the training field, there is insufficient evidence to support that complex training are superior in developing VJP than the combination of WT with PT in different sessions compound training.[141] Mihalik *et al.*[123] trained volleyball players twice per week, on Tuesdays and Thursdays, for 4 weeks. The complex group completed WT and PT on each day, with the compound group performing WT solely on Tuesdays and PT solely on Thursdays; both groups increased the vertical jump height 5.4 and 9.1% respectively. However, neither group improved significantly better than the other.

Effects of combined weight and plyometric training on vertical jump performance

The combination of WT and PT enhanced VJP (Table III) in untrained people by 4-17% [6, 8, 9, 15, 59, 61, 63, 64, 67, 131, 141-148] and trained subjects by 4-13%, [149, 150] there is a study that did not show any improvement.[16] Fatouros *et al.*[15] reported that the combination of WT and PT produced greater improvements in VJP (14.6%) than WT (9.3%) or PT (11.3%) alone. Adams *et al.*[6] also found that the combination

TABLE IV.—*Improvement in vertical jump performance with whole-body vibration training (VT).*

References	Training				Participants			
	Type	We	Se	Sets	N	Ge	Type	yr
							Untrained, Students, Recreational	
Bosco *et al.*[72]	VT	10 times		5-5	14	M	Students	25
Bosco *et al.*[135]	VT	10 days		5	7		Physically active subjects	20
Colson *et al.*[75]	VT+BaT	4	3		10	M-F	Regional basketball players	20
Delecluse *et al.*[78]	VT	12	3	1-3	18	F	Untrained	22
Roelants *et al.*[81]	VT	24	3	1-3	24	F	Untrained	65
Ronnestad *et al.*[134]	VT-S	5	2-3	3-4	7	M	Recreationally resistance-trained	21-40
Torvinen *et al.*[82]	VT	4-min			16	M-F	Untrained	24-33
Torvinen *et al.*[83]	VT	4-month	3-5		26	M-F	Nonathletic	23
Torvinen *et al.*[84]	VT	8-month	3-5		27	M-F	Nonathletic	23
							Trained, Elite, Athletes	
Annino *et al.*[70]	VT+BalT	8	3		11	F	Well-trained ballerinas	21
Cochrane *et al.*[74]	VT	5-min			18	F	Elite field hockey	22
Fagnani *et al.*[79]	VT	8	3	3-4	13	F	Competitive athletes	21-27

BaT: basketball training; VT-S: squat whole body vibration training; BalT: ballet training; We: weeks; Se: sessions per week; N.: number of subjects; Ge: Gender; F: female; M: male; Yr: years; SJ: squat jump; CMJ: countermovement jump; DJ: drop jump; V: other kind of jump; A: acceleration of the platform.

of WT and PT achieved greater improvement in vertical jump (10.7 cm) than WT (3.3 cm) or PT (3.8 cm) alone. Fowler *et al.*[125] observed that WT alone did not improve the CMJ height, while the combined group improved 6% the CMJ height.

Whole body vibration training

The vibration is an oscillatory motion that involves biomechanical and physiological processes that could improve muscle strength and power. The most common form of VT is performed with the subject standing on a vertical vibrating platform that primarily oscillate vertically or on a tilting vibrating platform that oscillate about a horizontal anteroposterior central axis.[151] The vibration frequency used to be between 25 and 50 Hz. The intensity of the vibration is determined by the amplitude of the vibration (the displacement of the oscillatory action, measured in mm), the frequency of the vibration, and the acceleration of the vibration (considered in g that represent the Earth's gravitational acceleration). During VT skeletal muscles of the lower extremities undergo small changes in muscle length which activate the stretch reflex loop.[81] Vibrations of muscles at rest elicit a tonic vibration reflex,[152] which includes activation of muscle spindles and subsequent excitation of α-motor neurons by Ia afferents and contraction of muscle fibers.[153]

The VT may elicit improvements in VJP by eliciting morphological, neural and hormonal adaptations. The vibration training enlarges the acceleration load of the subject that is exposed on a vibratory platform,[66] as muscle activation and load is increased it could cause some muscular adaptation as a muscle hypertrophy or changes in muscle fiber composition, however the changes in muscle CSA and fiber composition has not been investigated after a vibration training.[154] In a recent study, nineteen recreationally trained college-aged males were randomly assigned to a VT or a sham group.[155] The VT group completed a series of static, body weight squats on a vibrating platform at 30 Hz and amplitude of ~3.5 mm (vertical), whereas the control group performed the same series of exercises but without vibration. The ballistic isometric MVC of the triceps surae muscle complex was enhanced immediately after the VT (9.4%) and eight minutes after the end of the VT (10.4%), while not significant effects were observed in the control group. Neither the alpha-motoneuron excitability, assessed by analyzing the H-reflex, nor the EMG activity was altered by the VT. McBride *et al.* concluded that VT may increase MVC without eliciting changes in muscle activation or motoneuron ex-

Gain								Vibration		
Cm				%				Amplitude	Frequency	A
SJ	CMJ	DJ	V	SJ	CMJ	DJ	V	mm	Hz	g
	1.4				3.9			4	26	17
	0.6							10	26	54
				6.7				4	40	
					7.6			2.5-5	35-40	2.3-5.1
					19.4			2.5-5	35-40	2.3-2.7
	3.2				8.8				40	
	0.7				2.5			10	15-30	3.5-14
	2.5				8.5			2	25-40	2.5-6.4
	2.1				7.8			2	25-45	
	1.8				6.8				30	5
							8.1	6	26	
	2.8				9.6			4	35	17

citability.[155] The endocrine response to VT has been scarcely studied. Some studies observed increases in plasma growth hormone concentration after VT,[62, 72] whiles others did not find such an effect.[156] It has been reported that VT alone does not influence plasma testosterone concentration [62, 156] or only elicits small increases.[72] The mechanism by which VT could improve jumping ability is not clear and more research is required.

Effects of whole body vibration on vertical jump performance

Some authors have found an increase in VJP with the use of VT (Table IV) in untrained people from 3 to 19% [72, 75, 78, 81, 83-85, 157, 158] and trained people from 7% to 10%,[70, 74, 79] while other did not report such an effect.[62, 71, 73, 76, 77, 80, 82, 158, 159]

Delecluse *et al.*[78] reported that VT three times per week during 12 weeks elicited an 8 % increase in CMJ in untrained female, who also improved isometric (17%) and dynamic knee-extensor strength (9%) although no effect on the maximal speed movement was achieved. Torvinen *et al.*[83] concluded that four-month VT induced an 9 % improvement in the VJP which was accompanied by 4% enhancement of lower leg maximal isometric extension strength.

It has been suggested that the benefits of VT could be superior if VT is combined with WT.[157] Ronnestad [134] observed a superiority of squats performed on a vibration platform compared with squats without vibrations regarding maximal strength and explosive power as long as the external load is similar in recreationally resistance-trained men. In contrast, no differences in SJ performance were reported after a 6-week squat training program between the subjects performing only squats and those performing the same squat set with additional VT during the recovery periods.[159]

VT may also be a potential warm up procedure to increase VJP.[160] Cormie *et al.*[137] observed that a single bout of VT resulted in a significant higher jump height during the CMJ immediately following the vibration. However more research is required to clarify the optimal VT to improve VJP.

In contrast, De Ruiter *et al.*[76] reported that 11-week of VT did not improve CMJ performance and functional knee extensor muscle strength in young subjects. Similarly, Torvinen *et al.*[85] observed that an stimulus of 4-min of vibration was not sufficient to enhance the performance in several tests including vertical jump. Kvorning *et al.*[62] did not observed any improvement in VJP after 9-week training of VT, however a tendency towards an increase (P=0.052)

TABLE V.—*Improvement in vertical jump performance with electromyostimulation training (ET).*

Referentes	Training				–	Participants		
	Type	we	se	sets	N	ge	type	yr
							Untrained, Students, Recreational	
Deley *et al.* [90]	ET+GyT	6	3-1		8	F	Gymnastics national or regional level	12
Herrero *et al.*[18]	ET+PT	4	4		11	M	Students	21
Maffiuletti *et al.*[93]	ET+PT	4	3		10	M	Regional volleyball players	22
Malatesta *et al.*[94]	ET+VoT	4	3		12	M	Regional volleyball players	17
Paillard *et al.*[95]	ET	5	3		9 10	M	Trained students	18-33
							Trained, Elite, Athletes	
Babault *et al.*[87]	ET+RuT	12	3-1	1	15	M	Elite rugby players	22
Maffiuletti *et al.*[91]	ET+TeT	3	3		12	F-M	Well-trained tennis players	23
Maffiuletti *et al.*[92]	ET+BaT	4	3	12	10	M	Trained Basketball players	25

GyT: gymnastic training; PT: plyometric training; VoT: volleyball training; RuT: rugby training; TeT: tennis training; BaT: basketball training; we: weeks; Se: sessions per week; N.: number of subjects; Ge: gender; F: female; M: male; yr: years; SJ: squat jump; CMJ: countermovement jump; DJ: drop jump; V: other kind of jump; CMJs: 15 seconds of consecutive CMJ; Gain after: gain after training program

was observed in the combined VT and WT. The large inter-subject variance in the response to VT could explain the contradictories results.[154] It seems that the potential benefits from VT are higher for untrained than physically active subjects.[161]

Electromyostimulation training

The muscle can be excited by the nervous system or by an external electric current.[162] ET consists of passing an electric current across the muscle to induce a muscle contraction. ET has been used in rehabilitation.[163] and as a training method to improve physical performance.[164]

ET alone can increase muscle mass and strength,[165] however ET can not improve coordination between agonist and antagonist muscles,[166] as voluntary contractions can do, hence ET alone does not facilitate learning the specific coordination of complex movement like vertical jump. ET and voluntary muscle contraction are different modes of muscle activation and, therefore, induce different physiological adaptations.[166] This is likely the main reason why ET is used in combination with another training method to produce higher performance.[87, 89, 92, 94, 167]

Some of the mechanisms that could explain improvements in VJP with EMS include morphological, neural and biomechanical adaptations. The main morphological adaptation elicited by EMS that could potentially contribute to improve VJP is muscle hypertrophy. The CSA of the vastus lateralis (VL), vastus medialis (VM) and vastus intermedius muscles significantly increased (6%) after 8-week of EMS training.[168] Hypertrophy of the left quadriceps femoris with electrical stimulation reflected growth of each of the four quadriceps muscles.[169] Herrero *et al.*[18] observed a significant increase in CSA after EMS alone (9%) or combined with PT (7.1%). It has been also reported that EMS may induce neural adaptations.[168, 170] Gondin *et al.* observed that 8-week of EMS training significantly increased knee extensor MVC (27%) and was accompanied by a significant increases in muscle activation (6%), there was a significant increases in normalized EMG activity of both VL and VM muscles (69 and 39%, respectively) but not of rectus femoris muscle.[168] In another study isometric EMS training over 4 week significantly increased the plantar flexor voluntary torque under isometric conditions at the training angle (+8.1%) and the eccentric velocities torques (10.8-13.1%). The torque gains were accompanied by higher normalized soleus EMG activity and, in the case of eccentric contractions, also by higher gastrocnemii EMG.[170] The EMS may also induce biomechanical changes, like an increase of the pennation angle (14%),[168] the angle may affect negatively the VJP. The pennation angle of a muscle represents the position of muscle fibres in relation to connective tissue and tendon.[125] This angle determines the force transmission to the

Gain									Gain after		Electromyostimulation		
Cm				%					in %		intensity	duration	Pulse
SJ	CMJ	DJ	V	SJ	CMJ	CMJs	DJ	V	SJ	CMJ	mA	minutes	Hz
				>20.9	10.1						65-120	20	75
				7.5	7.3						0-120	29	120
				21							60-120	16	115-120
						4			6.5	5.4	0-100	12	105-120
5											0-120	15	80
3												60	25
3.2		2.1		10			6.6				0-100	12	100
										5.3	0-120	10	85
6.2				14						17	0-100	16	100

tendons and bones,[171] decreasing the amount of force transmitted in the line of action of the tendon as the pennation angle increases. However, it should be taken into consideration that this is commonly associated to muscle hypertrophy, independently of the mechanisms that elicited the increase in muscle size.

Effects of electromyostimulation on vertical jump performance

Some studies did not find any improvement on VJP.[88, 89, 172, 173] In many studies ET has been effective to improve vertical jump (Table V) in untrained subjects between 7% to 21%,[18, 90, 93-95] or trained people a 7% to 14%.[87, 91, 92]

The improvements in VJP have been observed when ET was combined with PT or sport practice,[18, 87, 90-94] except in one.[95]

Maffiuletti *et al.*[92] observed a significant increase in SJ (14%) in 10 trained basketball players after 4-weeks of ET of the knee extensors, 3 times a week, the ET consited of twelve 16-minute sessions, the intensity had a range between 0-100 mA and the pulse was 100 Hz. The improvement in vertical jump was accompanied by a significant increase in isokinetic strength at high velocities and in isometric strength at the angles adjacent to the training angle. At week 8, gains in isometric, isokinetic and SJ performance were maintained and the CMJ increased significantly. Another study was carried out on regional volleyball players.[93] ET combined with PT during 4 weeks improved SJ (21%) and CMJ (9%); the training sessions were carried out 3 times weekly, each session consisted of ET of the knee extensor muscles, 48 contractions, ET of the plantar flexor muscles, 30 contractions, and 50 plyometric jumps, the intensity range of the ET was 0-120 mA and the rectangular-wave pulsed currents 115-120 Hz. In this case, there was an augmentation of the knee extensors and plantar flexors maximal strength at week 2. The significant increases in maximal and explosive strength were maintained after 2 additional weeks of volleyball training. In contrast, using the same types of players, 4 weeks of ET, 3 times a week, plus volleyball practise was not enough to increases SJ or CMJ, in spite of a significant increase in the mean height and the power during a test consisting on 15 seconds of consecutive CMJs (4%),[94] however, jump height increased significantly in SJ (6.5%) and CMJ (5.4%) ten days after the end of the program training, the ET sessions lasted around 12 minutes, the intensity range was 0-100 mA and the pulse was 105-120 Hz [94] According to Herrero *et al.*[18] ET alone does not improve VJP, however when ET was combined with PT during 4-week it can increase the jumping height, SJ (7.5%) and CMJ (7.3%), in physical active men. The combined ET and PT also improved sprint time (-2.3%), maximal isometric strength (16.3%) and muscle CSA (7.1%). The ET took place 4 days a week during 4 weeks, each session lasted 34-minutes approximately; the frequency was 120 Hz and the intensity 0-120 mA. Babault *et al.*[87] analyzed the effect of 12-week ET on elite rugby players; ET was conducted on the knee extensor, plantar flexor, and gluteus muscles. The first 6 weeks consisted of 18 ET sessions, 12 minutes for each muscle group, with 3 sessions a week. During the remaining 6 weeks the ET was composed of only 1 ET session per week. The rectangular-wave pulsed current was 100 Hz, while the intensity range 0-100 mA. The SJ and DJ increased significantly 10% and 7%, respectively at the end of the training period, it was accompanied by a significant increased in maximal eccentric and concentric torque and squat strength.

Brocherie *et al.*[89] did not find any improvement in VJP with ET, they examined the influence of a short-term ET program on the strength of knee extensors, skating, and VJP of a group of ice hockey players. ET sessions consisted of 30 contractions (4-s duration, 85 Hz) and were carried out 3 times per week during 3-week. The isokinetic torque increased significantly in eccentric and concentric conditions as well as 10-m skating performance, however VJP decreased significantly for SJ (-2.9 cm), CMJ (-2.1 cm), and DJ (-1.3 cm).

Conclusions

VJP is important for success in many sports and may be used as a test to assess muscle power. Several training methods have been used to improve VJP; the most commonly used is PT and WT. More recently VT and ET have also been applied successfully to improve VJP. Since the vertical jump is a complex polyarticular dynamic movement that require intermuscular coordination any training program designed to improve VJP should include jumps to optimize coordination.

It is not clear which is the best training method to enhance VJP, although scientific evidence points to a combination of PT with WT. The PT enhances muscular force, RFD, muscular power and muscle contraction velocity. These effects have been shown in children, physically active subjects and elite athletes. It has been shown that PT enhances the mean CSA of type I, IIa, and IIa/IIx, the unloaded shortening velocity and peak power in all muscle fibers, with type II fibers showing the greatest improvement, without a significant enhancement of the specific isometric tension (force/cross-sectional area). PT may also enhance muscle stiffness allowing greater storage and release of elastic energy. Although irrefutable experimental evidence is lacking, it has been suggested that PT may also elicit increased inhibition of antagonist muscles after training, better co-contraction or increased activation of synergistic muscles, reduction of neural inhibitory mechanisms and increased agonist motoneuron excitability and synchronisation.

WT has been associated to improvements in force, velocity, power output, and RFD during jumping on a force plate, due to muscle hypertrophy, increased expression of MHC IIa, reduced expression of MHC IIx, as well as neural adaptations (increases in motor unit activation, firing frequency, synchronization of motor units, coordination, agonist activation and decreases in antagonist co-activation). High-speed, light loaded (30-40% 1RM), WT appear to be more effective than heavy-loaded (80 to 100% of 1RM) to improve VJP. Although WT at intensities as low as 16% 1RM is able to elicit some muscle hypertrophy, for a given training volume, muscle hypertrophy is more easily elicited by heavy-loaded than by light-loaded WT. In contrast, training-elicited changes in MHC appear to be less dependent on training intensity.

It seems that methods combining PT and WT are the most effective to enhance VJP, likely because they take advantage of the enhancement of maximal dynamic force through WT and the positive effects of PT on speed and force of muscle contraction through its specific effect on type II fibers and high-threshold alpha-motoneurones. There is strong evidence suggesting that a higher level of muscle hypertrophy can be achieved by combining WT and plyometric or eccentric exercises.

Although some investigations report increases in VJP with the use of VT particularly when combined with WT, lack of appropriate control groups in most of these studies, together with some negative findings in others, preclude a definitive conclusion of the effectiveness of these methods to improve VJP. In addition, it remains unknown by which mechanisms VT could enhance VJP.

ET has been shown to elicit muscle hypertrophy The VJP may be improved when ET is applied concomitantly with PT or practice of sports. Like with

VT, lack of appropriate control groups in most of these studies, together with some negative findings in others raises some doubts about the real contribution that adding ET to a training program may offer to enhance VJP.

Based on the results, coaches that want to improve VJP should known that PT, WT, VT and ET are effective methods. The combination of PT with WT seem to achieve a higher improvement than these two methods applied alone, the VT can also be used as a warm up procedure to increase vertical jump, and the ET should be add to the sport training or PT. Others aspects like materials, time for training, economic possibilities, situations can be taking into account in order to choose the most adequate method. There are still many questions that need to be resolved, especially in regard with the physiological mechanisms that explain the results obtained with specific training protocols. It is important what kind of neuromuscular adaptation may be elicited, and how should these training programs be inserted in the busy schedule of the elite athletes. More research should be undertake to decipher how diet and other concurrent training activities, like endurance or sprint training, may influence on the responses to vertical jump training. More research is also needed to establish criteria to choose the appropriate training loads and how to combine methods in the most efficient way.

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